



PROFILE

Mechanical Engineering graduate specialising in energy systems, hydrogen technologies, and control systems. Skilled in MATLAB, Simulink, FEA, CFD, and CAD, with a strong foundation in system modelling, simulation, and optimisation. Demonstrated ability to deliver measurable engineering improvements through applied design and analysis.

CONTACT

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- [joshua-srf.github.io](https://github.com/joshua-srf)

TECHNICAL SKILLS

- Simulation & Analysis
 - FEA & CFD (ANSYS, SolidWorks)
 - Structural, thermal & flow modelling
- Programming & Control
 - MATLAB & Simulink
 - Python & mathematical modelling
 - PID control & feedback systems
- Design & Manufacturing
 - CAD & CNC programming
 - SolidWorks (3D modelling & assembly)
- Energy & Thermofluids
 - Thermodynamics & heat transfer
 - Green hydrogen & electrolysis (AWE/PEM)
 - Fluid mechanics & system optimisation

DIGITAL & COMMUNICATION

- Adobe Illustrator & Photoshop
- Technical documentation & reporting
- Microsoft Office Suite

PROFESSIONAL SKILLS

- Analytical problem-solving
- Project planning & delivery
- Cross-functional team leadership
- Technical writing & presentation

LANGUAGES

- English (UKVI IELTS 8.0)
- Sinhala (Native)

JOSHUA FERNANDO

Mechanical Engineer • BEng (Hons)

EDUCATION

– BEng (Hons) Mechanical Engineering — Grade: 2:1

University of the West of England, Bristol, United Kingdom | Nov 2025

- Ranked 1st in cohort for Engineering Mathematics (98%); 2nd overall in Year 1 — awarded merit-based scholarship for exceptional academic performance

Modules: Thermodynamics | Fluid Mechanics | Applied Thermo-fluids | Dynamics | Vibrational Dynamics | Control Engineering | Structural Mechanics | Materials & Structures | Advanced Manufacturing | System Design

– Cambridge A-Levels — Physics | Mathematics | Chemistry

Lyceum International School, Sri Lanka | 2017 – 2019

WORK EXPERIENCE

– Engineering Apprentice — U.S.S Engineering (Huawei Partnership)

Colombo, Sri Lanka | Aug 2018 – Apr 2019

- Deployed fibre-optic infrastructure across rural regions; optimised cable routing through systematic site assessments, reducing installation rework
- Maintained QA compliance with international telecom standards, developing strong technical communication and field problem-solving skills

– STEM Tutor & Ambassador — CTC (Creative Tuition Collective)

Bristol, United Kingdom | Aug 2023 – Mar 2024

- Delivered Engineering and Mathematics tuition to 17–22 year olds across online and in-person settings, achieving consistent improvements in student performance
- Conducted STEM outreach sessions in Bristol schools as official Ambassador, promoting engineering as a career to secondary school pupils

– Social Media Assistant / Graphic Designer — MOD'Art University

Colombo, Sri Lanka | Oct 2020 – Aug 2021

- Produced technical graphics, marketing content and visual assets using Adobe Illustrator and Photoshop; managed digital platforms and advertising campaigns

– Activity Instructor — Bristol Activity Centre

Bristol, United Kingdom | Apr 2024 – Jul 2025

- Led structured activity sessions for diverse groups; enforced safety protocols and risk management procedures across all activities consistently

KEY PROJECTS

– Dissertation: Solar-Powered Green Hydrogen Generation

- Designed and modelled AWE & PEM electrolysis systems integrated with solar PV using MATLAB and FEA; optimised electrode geometry and system configuration to maximise hydrogen yield
- Engineered a 3D-printed gyroid electrode achieving a significant increase in hydrogen output efficiency; evaluated energy recovery pathways via HICEs, combustion engines, and fuel cells

– PID-Controlled Robotic Arm — AL5D Golf Placement System

- Designed and implemented a full PID control system with real-time sensor feedback for an AL5D robotic arm; achieved precise, repeatable ball placement with minimal positional error

– Sustainable Transport — DIY Electric Bike Conversion Kit

- Led multidisciplinary team to engineer an EV conversion kit and public charging network; performed full feasibility analysis and energy demand modelling to validate system viability

References available on request